

Speed, Distance & Acceleration

Quick reference sheet · Physics & Maths · keep handy for revision

1. Scalars and vectors

Some quantities need a direction to be complete, and some don't.

Distance — how far something moves. No direction needed — a **scalar** quantity.

Displacement — distance in a straight line from start to finish, *and* the direction of that line. A **vector** quantity.

Speed — how fast something moves. No direction needed — a **scalar** quantity.

Velocity — speed in a given direction. A **vector** quantity.

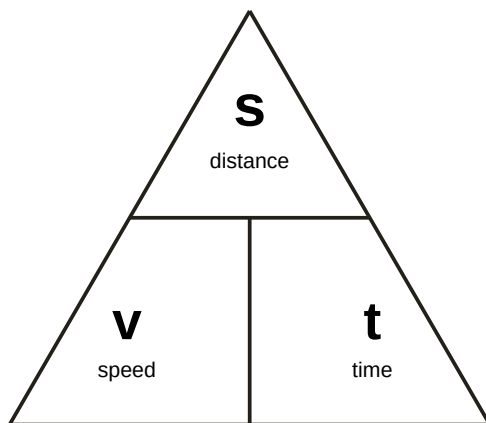
Typical speeds (worth memorising):

Walking	Running	Cycling	Sound in air
1.5 m/s	3 m/s	6 m/s	330 m/s

2. Distance, speed and time

distance = speed × time or $s = vt$

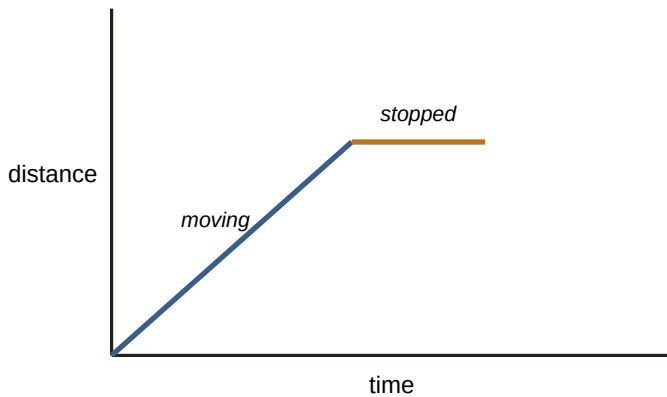
s — distance, in metres, m · v — speed, in metres per second, m/s · t — time, in seconds, s



Cover the letter you want to find. Top-over-bottom means divide, side-by-side on the bottom means multiply.

Cover $s \rightarrow v$ next to $t \rightarrow s = vt$. Cover $v \rightarrow s$ over $t \rightarrow v = s / t$. Cover $t \rightarrow s$ over $v \rightarrow t = s / v$.

3. Distance–time graphs

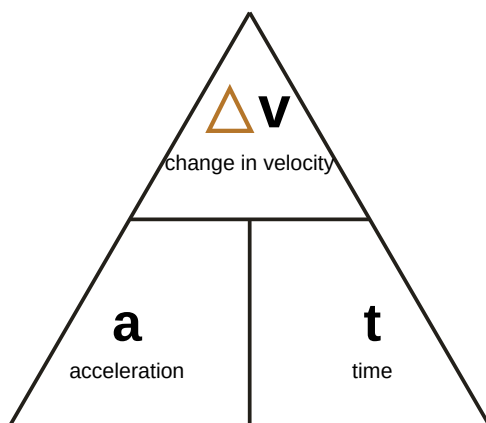


The **gradient** (steepness) of a distance–time graph is equal to the **speed**. A steeper line means a faster speed. A **flat, horizontal** line means the object has stopped (zero speed).

4. Acceleration

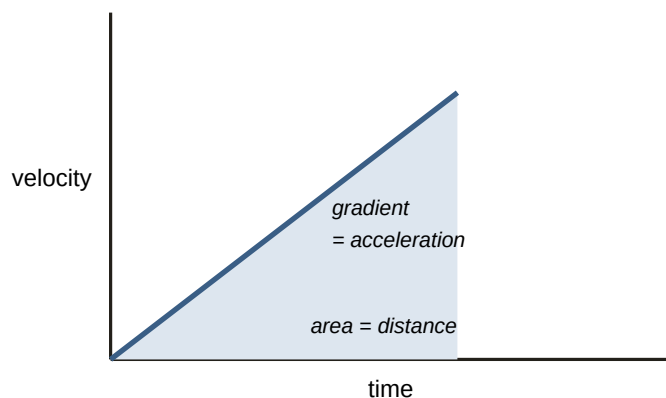
acceleration = change in velocity ÷ time or $a = \Delta v / t$

a — acceleration, in metres per second squared, m/s^2 · Δv — change in velocity, in m/s · t — time, in s



A negative acceleration means the object is **decelerating** (slowing down). Near the Earth's surface, an object in free fall accelerates at about **9.8 m/s^2** .

5. Velocity–time graphs



The **gradient** of a velocity–time graph is equal to the **acceleration**. The **area** underneath the graph (down to the time axis) is equal to the **distance travelled** — for straight-line graphs this is just the area of the triangle or rectangle formed.

6. Worked examples

Given	Find	Working	Answer
$v = 15 \text{ m/s}$, $t = 8 \text{ s}$	s	$s = vt = 15 \times 8$	120 m
$s = 180 \text{ m}$, $t = 12 \text{ s}$	v	$v = s / t = 180 / 12$	15 m/s
$s = 90 \text{ m}$, $v = 6 \text{ m/s}$	t	$t = s / v = 90 / 6$	15 s
$\Delta v = 18 \text{ m/s}$, $t = 6 \text{ s}$	a	$a = \Delta v / t = 18 / 6$	3 m/s^2
$a = 5 \text{ m/s}^2$, $t = 4 \text{ s}$	Δv	$\Delta v = at = 5 \times 4$	20 m/s